Translution-SNet: A Semisupervised Hyperspectral Image Stripe Noise Removal Based on Transformer and CNN

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***Abstract*— Hyperspectral remote sensing images (HSIs) have been applied in urban planning, environmental monitoring, and other fields. However, they are susceptible to noise interference, such as Gaussian noise, stripe, and mixed noises, from various factors in the imaging process, which greatly limits their applications. Although previous efforts to improve HSI quality have achieved remarkable results, there are still many challenges to be solved. To avoid the poor generalization ability and improve the stripe removal performance of the network in real scenarios, in this article, we proposed a novel deep learning model (Translution-SNet) for HSI stripe noise removal based on a semisupervised training strategy that applies a convolution and transformer for feature extraction. Moreover, we used an unbiased estimation method to calculate the loss function of the unsupervised part from noisy data without a clean image. The semisupervised method improved the ability of Translution-SNet to deal with various complex stripe noises during stripe removal and strengthened its robustness and generalization ability. Our experimental results showed that Translution-SNet could robustly handle stripe noise of images with different loads and achieve satisfactory results, proving its feasibility and effectiveness. In addition, Translution-SNet showed good generalization ability.**

***Index Terms*— Hyperspectral image destriping, remote sensing, semisupervised, transformer, unbiased estimate.**

NOMENCLATURE

CNN	Convolutional neural network.
CSWin	Cross-shaped window.
CCD	Charge-coupled devices.
HSI	Hyperspectral remote sensing image.
LRMR	Low-rank matrix recovery.
PCA	Principal component analysis.
RPCA	Robust principal component analysis.
SURE	Stein unbiased risk estimate.
ASSTV	Anisotropic spectral–spatial total variation.
LRTV	Total-variation-regularized low-rank matrix factorization.
LRTF_DFR	Double-factor-regularized low-rank tensor factorization.
NMoG_RPCA	Non-i.i.d. mixture of Gaussians robust principal component analysis.
HSI-DeNet	Hyperspectral image restoration via convolutional neural.
SGIDN	Satellite-ground integrated destriping network.
SWin	Shifted windows.
MAD	Median absolute deviation.

Manuscript received 16 March 2022; revised 25 April 2022 and 20 May 2022; accepted 7 June 2022. Date of publication 11 July 2022; date of current version 18 July 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 41971412. *(Corresponding author: Yonghua Jiang.)*

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Digital Object Identifier 10.1109/TGRS.2022.3182745

I. INTRODUCTION

COMPARED with natural images, HSIs have numerous spectrums, which can reflect different characteristics of ground objects and play important roles in growing national economies [1]–[3]. HSIs are precise because of their large number bands; however, they are easily interfered with sensor hardware and other factors, thus resulting in image quality issues and restricting their application [4], [5]. The costs of improving image quality from the hardware aspect are high and cannot fully solve the quality degradation issue. Therefore, efforts have been made to improve image quality through image processing. For HSI, random noise and stripe noise are the main factors that lead to image degradation. In this article, we focused on removing stripe noise.

A. Related Works

Stripe noise is formed by many factors and mainly stems from inconsistent responses between CCDs and register readout errors. Compared with random noise, stripe noise has obvious geometric and anisotropic characteristics. Although research into stripe noise removal started relatively recently, substantial information has been gathered and several methods have been developed. According to different structures and strategies, this article divides destriping methods into two categories: nondeep learning methods and deep learning methods.

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1) *Nondeep Learning Method*: This destriping models use the mechanism of stripe noise, prior knowledge, or stripe noise attributes. In general, nondeep learning methods use frequency-domain filtering, total variation, and low-rank sparse methods [6]–[8]. For example, the first stripe noise removal method, developed by Horn and Woodham [9], was based on histogram matching for Landsat satellite stripe noise removal. Corsini *et al.* [10] designed a statistical method for MOS-B stripe noise removal by estimating the homogeneous region equalization curve for stripe image calibration.
   Chen and Qian [11] combined PCA and wavelet transform to remove stripe noise. Liu *et al.* [12] proposed a destriping method based on 3-D wavelet transform combined with low-rank prior knowledge of images to process stripe noise in push-broom imaging. Chang *et al.* [13] regarded HSIs as the spectral–spatial volume and proposed an anisotropic spatial–spectral total variational regularization stripe removal method to further improve the destriping effect. Based on RPCA [14], Zhang *et al.* [15] developed a method of multitype noise removal based on LRMR, which achieved the denoising effect by optimizing the blocks with overlapping regions. Based on the LRMR method, He *et al.* [16] used global spatial–spectral total variation as a regular term to avoid over-smoothing in the spatial and spectral dimensions of the reconstructed image to improve the effect of image reconstruction.
2) *Deep Learning Methods*: This type of methods achieves destriping by designing a neural network. Over the past decade, deep learning methods have been successfully applied in the field of remote sensing image processing for object classification, change detection, target recognition, and semantic segmentation [17]–[21]. In terms of natural image denoising, Denoising Convolutional Neural Network (DnCNN) [22], FFDNet [23], and CBDNet [24] all show certain advantages, which provide the basis and ideas for HSI data denoising and destriping. Presently, most HSI data destriping or denoising methods based on deep learning methods use paired data comprising simulated stripe noise and clean images for network training. For example, Chang *et al.* [25] proposed the HSI-DeNet, which employed residual structure and atrous convolution, for HSI destriping. Zhang *et al.* [26] extracted the multiscale features of the noisy band and its adjacent bands and made full use of the spatial–spectral information of HSI data for hybrid noise removal. Sun *et al.* [27] proposed the multiscale low-rank deep back projection fusion network (MLR-DBPFN) to obtain high spatial resolution hyperspectral (HR HS) data, which can suppress the noisy spectrum and improve the noisy HS fusion performance.

There are also destriping methods that combine image prior knowledge with deep neural networks. For example, Zhong *et al.* [28] used the gradient information of images as input data to train a destriping model. Chang *et al.* [29] combined discrete wavelet transform with convolutional feature extraction for HSI data stripe removal. The attention mechanism is one of the commonly used methods in HSI data destriping methods. Shi *et al.* [30] enhanced denoising performance by using 3-D spatial and spectral attention. Wang *et al.* [31] improved destriping performance by applying cross channel enhanced spatial and spectral attention to make their model focus on key information. In addition, based on FFDNet, Maffei *et al.* [32] used a noise map together with the noisy band and its adjacent bands as the network inputs, which enabled the model to obtain the ability of blind denoising while utilizing spatial and spectral information. Yuan *et al.* [33] designed a noise level estimation subnetwork to estimate the noise level in the image and used the estimated noise level to train the denoising network.

Some scholars have tried to apply real stripe-contained data to network training to improve the stripe removal performance of models in real scenarios. For example, Wang *et al.* [34] proposed a self-supervised HSI denoising network that generated noise from real noise-contained data; although the real data were used to generate noise, it did not directly participate in training the denoising network. Song *et al.* [35] designed an unsupervised HSI data stripe removal network that combined wavelet transform and CycleGAN. Nguyen *et al.* [36] proposed an unsupervised denoising network based on SURE [37].

In general, nondeep learning methods can flexibly apply prior knowledge and do not require a large amount of paired training data; however, they do require complex models, which are difficult to optimize. In contrast, deep learning-based methods have powerful feature learning and feature modeling capabilities; however, their generalization ability is limited because of using simulated stripes for training data. Therefore, the question of how to use a large amount of real stripe-contained data to improve model generalization ability needs an urgent solution.

B. Vision Transformer

Transformer-based architectures were originally used for natural language processing. Transformers have good long-distance modeling and parallel feature extraction capabilities. Recent studies have shown that networks based on pure transformer architecture are equally effective and competitive in visual tasks. The vision transformer (ViT) is one of the representative works of using transformers for visual tasks. In ViT, images are divided into a series of patches (tokens) of the same size without overlapping areas. To model features, each token is given position information through position embedding and then enters the standard transformer layers of the fixed structure, which comprise multiheads self-attention (MHSA) and feed forward (FFD) modules. The transformer expresses the correlation of different regions of the image through the vector dot product operation.

Pure transformer structures can be comparable to CNNs in various tasks of computer vision [38]–[41]. For example, Liu *et al.* [42] proposed a transformer that builds a hierarchical structure using SWin, which achieved good results in object detection and semantic segmentation tasks and can thus be used as a backbone for vision tasks. Lin *et al.* [43] proposed a pyramid transformer with deformable attention for image processing.

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Presently, some scholars use transformer structures in remote sensing image processing tasks, such as Lei *et al.* [44], who used a transformer for the super-resolution reconstruction task of remote sensing images, exploiting different level features through the transformer-based structure. Hong *et al.* [45] proposed a transformer-based HSI data classification method using the link between HSI data bands. Dong *et al.* [46] explored a ViT structure for polarimetric synthetic aperture radar (PolSAR) image classification with both supervised and unsupervised learning. Chen *et al.* [47] used a transformer for urban change detection and used several semantic tokens to represent changing regions in bitemporal images. Zhang *et al.* [48] proposed a deep learning model combined transformer with CNN for semantic segmentation of very-high-resolution remote sensing imagery.

C. Contributions

In this article, we propose a semisupervised HSI stripe noise removal method that aggerates convolution feature extractor and transformer feature extractor to fully utilize the local and global features. In addition, to employ the real stripe-contained data to model training, we will employ the SURE approach to compute the loss function.

This article is novel for several reasons.

1) To avoid the poor generalization ability of destriping caused by the network overfitting to the simulated stripe noise distribution and to improve the stripe removal performance of the network in real scenarios, real data are used for network training in a semisupervised manner.
2) To calculate the loss function without the need for corresponding clean images in the real data in the training stage, this article adopts an unbiased estimation method based on SURE to estimate the mean square error (MSE) and replaces MSE loss with SURE loss.
3) To better combine local and global features, this article uses a transformer and convolution to extract global and local features, respectively, to enhance the stripe removal performance. In particular, we use a CSWin transformer to better capture the correlation between pixels from a global perspective to extract key features.

The remainder of this article is organized as follows. Section II describes problem formulation and the details of the proposed HSI stripe removal model; Section III presents and discusses the simulated and real-data experimental results; Section IV analyzes the parameters sensitivity and the effectiveness of the semisupervised training strategy; and finally, Section V presents the conclusions.

II. Materials and Methods

A. Problem Formulation

In general, HSI data are considered a $W \times H \times B$ 3-D cube, where W, H, and B are the height, width, and depth of the HSI data, respectively. The degrade model of the HSI data can be defined as

$$Y = X + N. \tag{1}$$

Obtaining stripe-free data X from Y degraded by stripe noise N can be seen as an ill-conditioned inversion process, and it should be noted that the intensity and type of stripe noise may be different in each band of HSI data.

B. CSWin Transformer

Transformers have natural advantages in global feature modeling; however, the spatial distribution of stripe noise is uncertain and local features are not enough to reflect the correlations between different pixels. Stripe noise has the characteristics of directionality and anisotropy, and the key features between stripe components and stripe-free components can be more accurately captured through global features. Therefore, we adopted a CSWin transformer [49] for feature extraction. The CSWin transformer can be considered an improvement to the SWin transformer.

In the SWin transformer, the receptive field of the token is limited and the window is gradually expanded by the stacking layer to improve the receptive field and obtain global information, increasing the computation amount. At the same time, in order to realize the information transfer between windows, SWin transformer adopts SWin to obtain the information between different windows. As shown in Fig. 1(a), the blue boxes are the local window, which are, respectively, represented as A, B, C. After the shift window operation, the positions of the three windows have changed, and then, the self-attention calculation of the local window is performed. To reduce the amount of computation while expanding the receptive field, the CSWin transformer uses a CSWin for feature extraction and performs self-attention feature extraction from the horizontal and vertical directions. Suppose the input feature map size is $H \times W \times C$, where H is the height, W is the width, C is the band and SW is the stripe width. Taking self-attention in the row direction as an example, a local window of size SW $\times$ H is obtained for the self-attention calculation. It is worth noting that the CSWin transformer divides the number of multiheads into two, which are used to calculate self-attention in the row and column directions, respectively. The calculation process of the CSWin transformer is shown in Fig. 1(b).

C. Proposed Method

Here, we propose a novel CNN-transformer-based method with a semisupervised scheme for HSI data stripe noise removal, named Translution-SNet. The flowchart of Translution-SNet is shown in Fig. 2.

While CNN possesses the ability of a powerful feature model, it is limited by the size of the receptive field and its ability to model global features is weak. In contrast, the transformer can model long-distance dependencies and capture global information. However, for the same model size, the transformer requires more training data than CNN to achieve the same effect. Therefore, we proposed Translution-SNet, which combines the advantages of CNN and transformer by adopting the dual-branch structure of transformer and CNN to extract features from local and global perspectives. Simultaneously, we aimed to reduce the loss of spectral information using skip connections.

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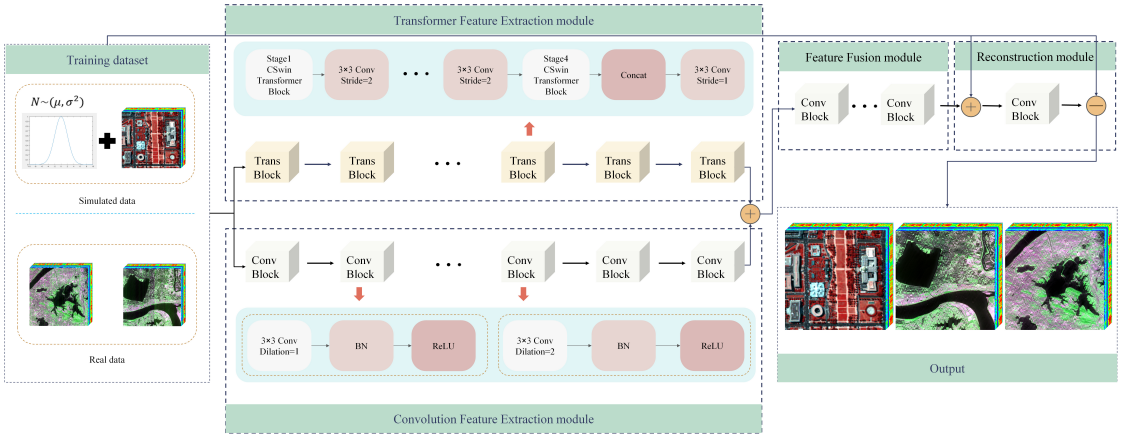


Fig. 1. Flowchart of the proposed Translution-SNet.

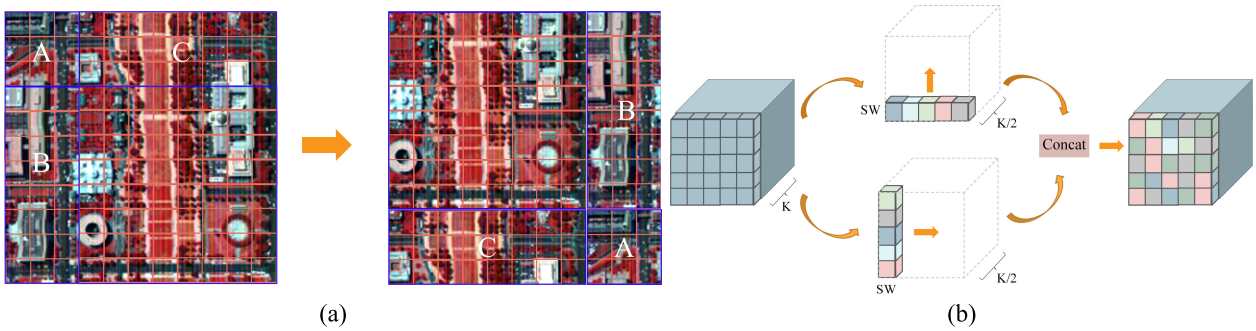


Fig. 2. (a) Self-attention in SWin transformer. (b) Self-attention in CSWin transformer. SW denotes the SWth local window. k denotes the number of MHSA.

The process was as follows: input data were extracted by the transformer and convolution feature extraction module, and then, the extracted feature feed was sent to the feature fusion module to be fused with global and local features before being sent to the reconstruction module to obtain the final output results.

1) Transformer Feature Extraction Module: The transformer feature extraction module proposed in this article (Fig. 3) contains a convolutional layer and four stages. The first convolution layer embeds the input image. Each stage comprises several CSWin transformer blocks and a convolutional layer with a convolution kernel size of 3×3 and stride $= 2$ is added between each stage to reduce the number of tokens. In addition, a skip connection is used to ensure that the image information remains intact. Finally, feature fusion is performed through a convolutional layer with a convolution kernel size of 3×3 and stride $= 1$. The CSWin transformer block is composed of layer normalization, CSWin self-attention, and multilayer perceptron. Moreover, two skip connections are used to reduce information loss. The detailed configurations of the transformer feature extraction module are summarized in Table I.

2) Convolution Feature Extraction Module: The convolution feature extraction branch comprises convolution, batch normalization, and rectified linear unit (ReLU) [convolution layer, batch normalization and ReLU (CBR)]. The architecture of convolution extraction model is shown in Table II. Here,

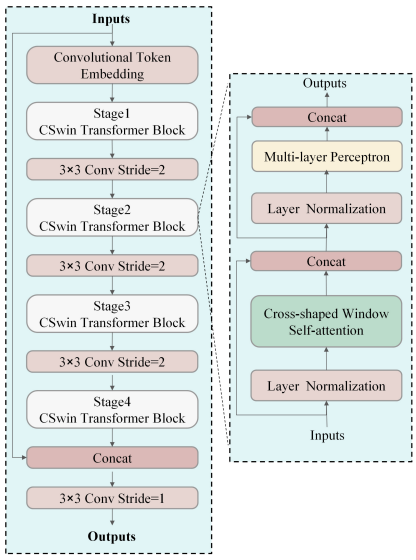


Fig. 3. Illustration of CSWin transformer: SW denotes the stripe width and k denotes the number of MHSA.

the size of all convolution kernels is 3×3 and the receptive field is increased by atrous convolution to obtain features at different scales. It is worth noting that we do not use a pooling layer. Many studies show that pooling operation will inevitably lead to the loss of image information for pixel-level low-level visual tasks [50], [51]. Simultaneously, we use skip connection to further avoid the loss of image information and

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TABLE I
DETAILED CONFIGURATIONS OF THE TRANSFORMER FEATURE EXTRACTION MODULE

Channels	Blocks in 4 stages	SW in 4 stages	Heads in 4 stages
64	[1,3,6,1]	[1,2,4,4]	[2,4,4,4]

TABLE II
ARCHITECTURE OF CONVOLUTION EXTRACTION MODEL

Layer	Kernel size	Filter number	Dilation	Image size
CBR1	3×3	64	1	40×40×64
CBR2	3×3	64	2	40×40×64
CBR3-CBR4	3×3	64	1	40×40×64
CBR5	3×3	64	2	40×40×64
CBR6-CBR8	3×3	64	1	40×40×64
CBR9	3×3	64	2	40×40×64
CBR10-CBR11	3×3	64	1	40×40×64
CBR12	3×3	64	2	40×40×64
CBR13-CBR15	3×3	64	1	40×40×64
Concat	/	/	/	40×40×74
Tanh	/	/	/	40×40×74
Convolution	3×3	64	1	40×40×64

increase training speed. In addition, the feature fusion and reconstruction modules are composed of CBR, with depths of 6 and 1, respectively.

3) Loss Function of Translution-SNet: We used a semisupervised strategy to train Translution-SNet. Therefore, the loss function consisted of two parts: supervised and unsupervised losses. The supervised loss function used the MSE and was defined as follows:

$$L_{\text{supervised}} = \frac{1}{2n}\sum_{i=1}^{n}\|X_i - R_i\|^2 \tag{2}$$

where n is the batch, X is the clean image, and R is the destriped image.

For the unsupervised loss function, there were no corresponding clean data, meaning that MSE could not be applied. Therefore, we introduced SURE to obtain an unbiased estimate of MSE [52], which can only obtain a noisy image. SURE can be described as follows:

$$L_{\text{unsupervised}} = \frac{1}{n}\|y - F(y)\|^2 - \sigma^2 + \frac{2\sigma^2}{n}\text{tr}\left(\frac{\partial \text{F}(y)}{\partial y^T}\right) \tag{3}$$

where y is the real noisy data, f is the Translution-SNet, and F(y) is the output of Translution-SNet.

The last term of (3) can be directly calculated under special conditions; however, for nonlinear estimators, such as deep learning methods, it needs to be estimated by Monte Carlo SURE. The specific calculation is as follows:

$$\text{tr}\left(\frac{\partial \text{F}(y)}{\partial y^T}\right) \approx b^T\left(\frac{\text{F}(y+\varepsilon b) - \text{F}(y)}{\varepsilon}\right) \tag{4}$$

where b is a Gaussian stripe that obeys the standard normal distribution and ε is a minimum value in the range of 10e-5–10e-2. In this article, ε = 10e-5.

For parameter σ, which denotes the noise intensity of the real stripe-contained data, we introduced the MAD estimator to estimate σ in the highest subbands (W) of the wavelet transform [53], as follows:

$$\sigma = \frac{\text{MAD}(|W|)}{0.6745}. \tag{5}$$

Subsequently, the unsupervised loss function can be calculated by SURE. Finally, the loss function for Translution-SNet training can be defined as

$$L = L_{\text{supervised}} + \alpha L_{\text{unsupervised}} \tag{6}$$

where α is the adjustment coefficient, set as 0.1.

III. EXPERIMENTAL RESULTS

A. Training Dataset and Parameters Setting

We choose the Washington DC Mall HSI dataset as the training data with a size of 1080 × 303 × 191, adding the simulated stripe noise to the clean image to form a degraded image for network training. Meanwhile, considering that Translution-SNet is a semisupervised network, we chose GaoFen-5 (GF-5) data disturbed by stripe noise as the training data for the unsupervised part. All training data were split into a size of 40 × 40 × 10, and data augmentation was performed on the training data. The network was implemented using the Pytorch framework with Python language. The learning rate was set at 0.00001 and the batch size was set at 40. We used Adam as the optimizer with the parameters $\beta 1$, $\beta 2$, and ε, which were set to 0.9, 0.999, and 1e-8, respectively.

B. Comparison Methods and Evaluation Index

To verify the effectiveness and competitiveness of our method, we used representative HSI data stripe noise removal methods: ASSTV [13], LRTV [54], LRTF_DFR [55], NMoG_RPCA [56], HSI-DeNet [25], and SGIDN [28] for comparison. Experiments were carried out on simulated and real data with multiple loads. Here, we used peak signal-to-noise ratio (PSNR), structural similarity index measurement (SSIM), and spectral angle map (SAM) as the indicators [57], [58].

C. Simulated Experimental Results

In the simulation experiment, we designed two different cases to verify and evaluate the destriping performance of Translution-SNet.

Case 1: The stripe noise intensity of all bands was different, and all columns of the image were added with a Gaussian stripe with a mean 0 and standard deviation range from 0 to 30.

Case 2: The stripe noise intensity of all bands was the same, and the image added a Gaussian stripe and deadlines with a mixed mean of 0 and a standard deviation of 20.

The index evaluation results of the simulation experiments are shown in Table III. Figs. 4 and 5 display the results with pseudo-colors (bands 57, 27, and 17).

Fig. 4 shows that Translution-SNet and the comparison methods had inhibitory effects on stripe for Case 1, but performance varied. After ASSTV processing [Fig. 4(c)], most of the stripe noise was suppressed and the spectral and structural

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TABLE III
EVALUATION INDEXES OF SIMULATED EXPERIMENTS

Case	Index	Method					
		ASSTV	LRTV	LRTF_DFR	NMoG_RPCA	HSI-DeNet	Translution-SNet
Case 1	MPSNR	31.41	28.54	28.06	30.89	34.82	35.57
	MSSIM	0.865	0.627	0.702	0.885	0.9033	0.9246
	MSAM	5.325	12.036	9.209	5.168	5.071	4.969
Case 2	MPSNR	32.03	29.26	27.65	30.78	33.96	34.84
	MSSIM	0.868	0.712	0.784	0.882	0.9003	0.9272
	MSAM	4.267	7.847	6.463	3.850	3.746	3.597

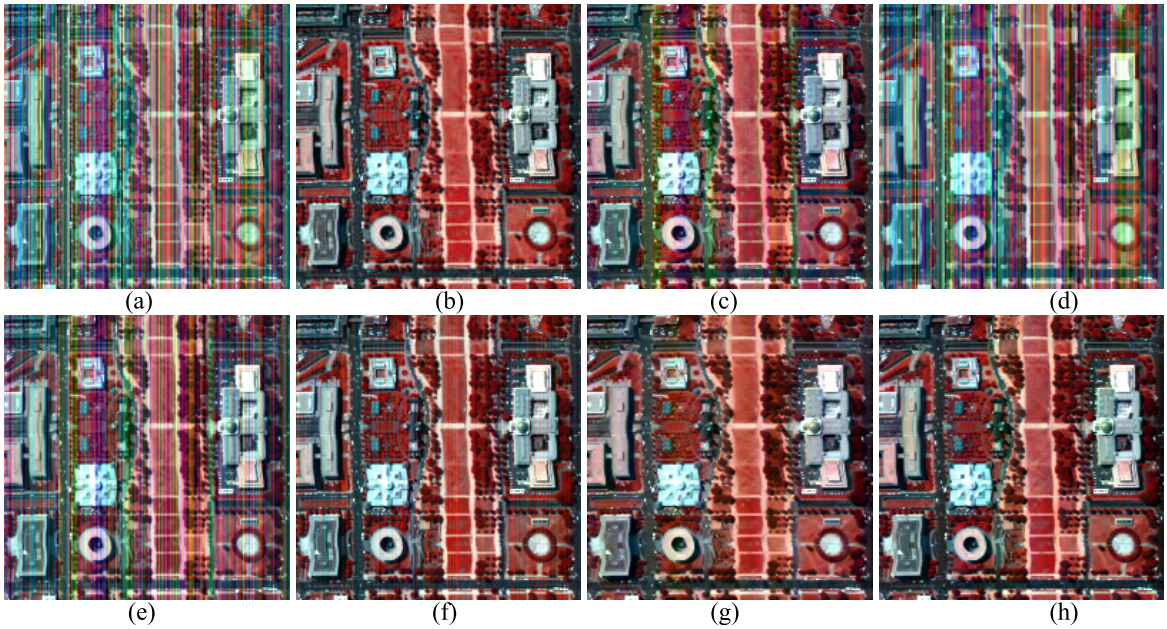


Fig. 4. Destriping results of Case 1: (a) degraded image; (b) clean image; (c) ASSTV; (d) LRTV; (e) LRTF_DFR; (f) NMoG_RPCA; (g) HSI-DeNet; and (h) Translution-SNet.

information of the original image were well-preserved, but there were still obvious stripe noise residues. The stripe noise processing ability of LRTV [Fig. 4(d)] was not good, and many stripes were not well-suppressed. This method also resulted in over-smoothing. LRTF_DFR [Fig. 4(e)] partially suppressed stripe noise but a large amount of stripe remained. For NMoG_RPCA [Fig. 4(f)], only a small amount of stripe noise was not removed; however, the spectral information of the image was lost.

For HSI-DeNet [Fig. 4(g)], the stripe noise was well-suppressed, leaving no obvious stripe noise residue. However, the structural and spectral information of the original image could not be accurately recovered and over-smoothing occurred. For the proposed Translution-SNet method [Fig. 4(h)], stripe noise was completely removed, the spectral and structure information of the restored images were completely preserved, and satisfactory destriping results were obtained.

The stripe noise mixed deadlines of Case 2 presents a greater challenge to destriping methods, as not only does strip noise need to be suppressed but pixels lost by deadlines need to be restored. From Fig. 6, after ASSTV processing [Fig. 5(c)], the restored image had no obvious stripe noise residue, but the restoration of pixels lost by deadlines was not ideal. In addition, there were deficiencies in the recovery of spectral information. Although LRTV and LRTF_DFR (Fig. 5(d) and (e), respectively) suppressed some stripe noises, there was still a lot of obvious stripe noises and the deadlines were not repaired. NMoG_RPCA [Fig. 5(f)] suppressed stripe noise well but noise was not completely removed and deadlines remained.

For HSI-DeNet [Fig. 5(g)], most of the stripe noise was well-suppressed and the deadlines were repaired; however, the repair trace was obvious. Translution-SNet [Fig. 5(h)] completely removed stripe noise and repaired deadlines to a satisfactory standard. Overall, the spectral and texture structural information were preserved and restored.

D. Real-Data Experimental Results

The main motivation of proposing Translution-SNet was to improve the generalization ability of the model. Therefore, experiments on real stripe-contained data of different satellites and loads were conducted to verify the generalization ability.

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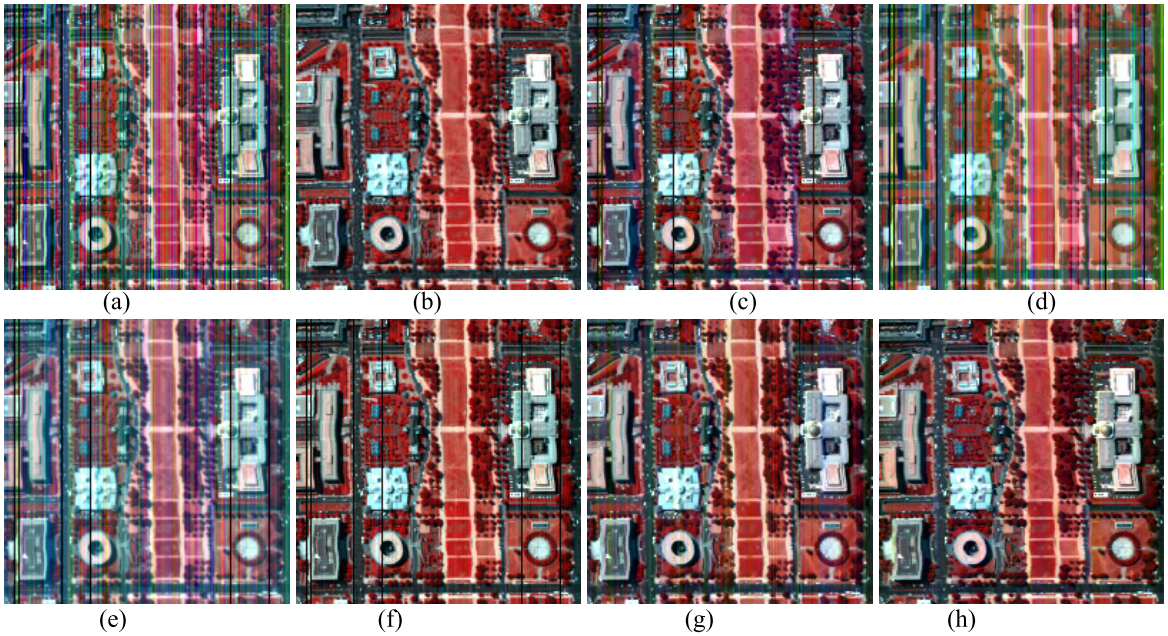


Fig. 5. Destriping results of Case 2: (a) degraded image; (b) clean image; (c) ASSTV; (d) LRTV; (e) LRTF_DFR; (f) NMoG_RPCA; (g) HSI-DeNet; and (h) Translution-SNet.

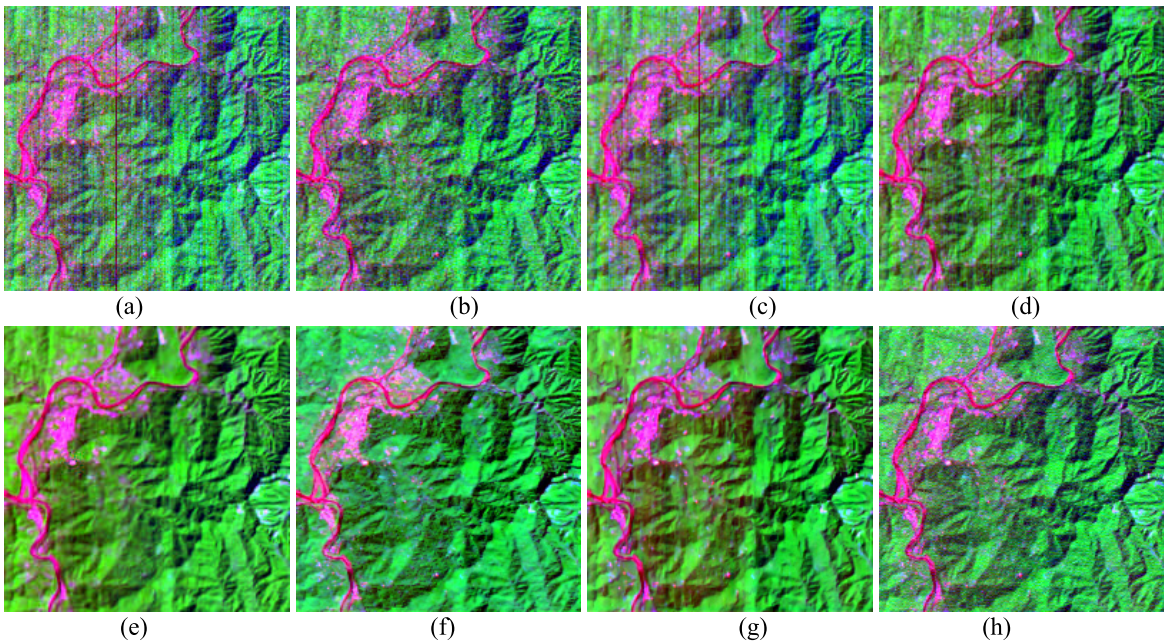


Fig. 6. Destriping results of EO-01 data: (a) degraded image; (b) ASSTV; (c) LRTV; (d) LRTF_DFR; (e) NMoG_RPCA; (f) HSI-DeNet; (g) SGIDN; and (h) Translution-SNet.

Here, we used Hyperion, MODIS, GF-5, Zhuhai-1, and Mars data. The results of the real-data experiments are displayed in Figs. 6–10.

1) Hyperion Dataset: EO-01 data are captured through the Hyperion sensor with size 200 × 200 × 210 and are mainly degraded by stripe, deadline, and Gaussian mixed noise. The destriping results are displayed in Fig. 6, with pseudo-color image combing bands 1, 73, and 132.

Fig. 7 shows that different methods had different degrees of processing effect on stripe noise: ASSTV [Fig. 6(b)] removed most stripe noise, but the restoration of deadlines was not thorough enough, with obvious restoration traces remaining. For LRTV [Fig. 6(c)], only part of the stripe noise was suppressed, deadlines remained, and over-smoothing occurred. For NMoG_RPCA [Fig. 6(e)], no obvious stripe noise or deadline residues remained; however, there were obvious spectral changes and loss of detail.

For HSI-DeNet and SGIDN (Fig. 6(f) and (g), respectively), large amounts of stripe noise and deadlines were well-suppressed and repaired, but the spectral information changed

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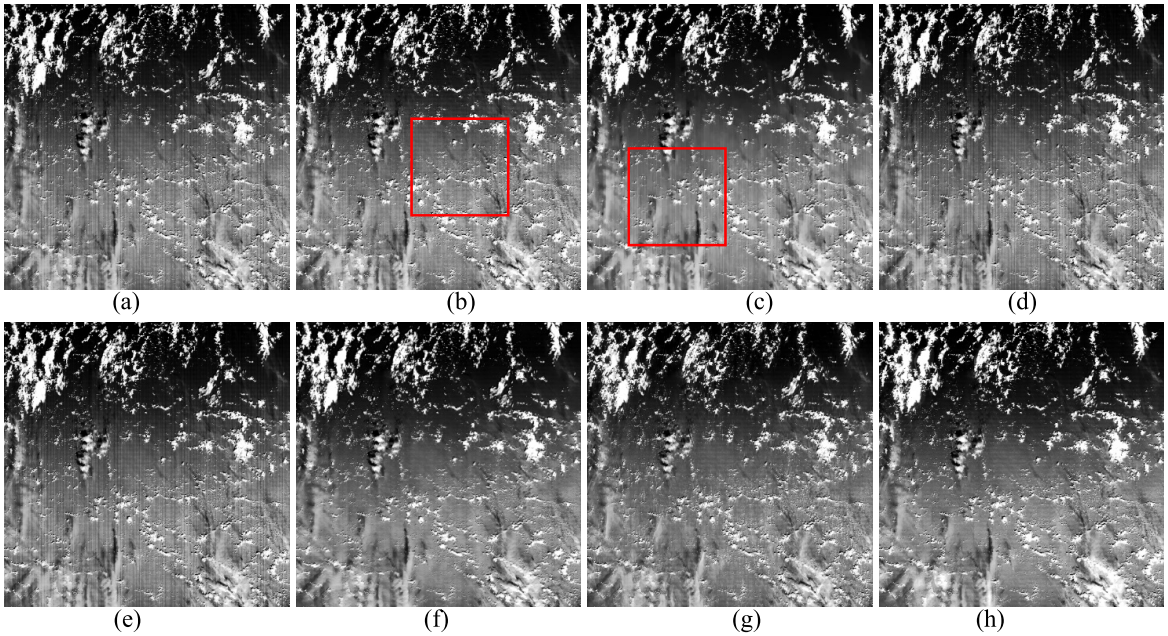


Fig. 7. Destriping results of MODIS data: (a) degraded image; (b) ASSTV; (c) LRTV; (d) LRTF_DFR; (e) NMoG_RPCA; (f) HSI-DeNet; (g) SGIDN; and (h) Translution-SNet.

and the detail of ground objects was lost. In addition, over-smoothing occurred. For Translution-SNet [Fig. 6(h)], the original spectral and feature information were preserved while restoring the image and satisfactory results were achieved in stripe removal.

2) MODIS Dataset: MODIS data with size 600 × 600 are mainly degraded by stripe. The destriping results are displayed in Fig. 8. For ASSTV [Fig. 7(b)], most stripe noise was suppressed but small amounts remained. LRTV [Fig. 7(c)] did not remove all stripe noise, and the details of the ground objects were blurred. For LRTF_DFR and NMoG_RPCA (Fig. 7(d) and (e), respectively), due to the limited processing effect of the stripe of the MODIS dataset, many stripe noise residues remained and the processing effect was insufficient. For HSI-DeNet and SGIDN [Fig. 7(f) and (g)], almost all stripe noise was suppressed. Translution-SNet [Fig. 7(h)] completely suppressed stripe while also outperforming the other methods in spectral information and feature detail retention.

3) GF-5 Dataset: GF-5 data are obtained by advanced hyperspectral imager (AHSI) with visible and near infrared (VNIR) and have a size of 200 × 200 × 150. GF-5 data are mainly severely disturbed by stripe noise. The destriping results are shown in Fig. 8, with bands 140, 1, and 60.

The comparison methods and Translution-SNet had inhibitory effects on stripe noise but the degree of inhibition and processing effect differed. As shown in Fig. 8(b), ASSTV suppressed most stripe noise, but was not thorough enough as obvious noise residue remained. Following LRTV and LRTF_DFR (Fig. 8(b) and (d), respectively), a small amount of stripe noise remained, ground object boundaries were blurred, and spectral information and image detail were lost. NMoG_RPCA, HSI-DeNet, and SGIDN have achieved a well-suppressed effect on stripe noise. The disadvantage is that after processing by NMoG_RPCA, HSI-DeNet, and SGIDN displayed different degrees of over-smoothing caused by the loss of ground details and spectral information, as shown in Fig. 8(e)–(g). In contrast, Translution-SNet [Fig. 8(h)] achieved a thorough suppression of stripe noise and better retained spectral information and feature detail.

4) Zhuhai-1 Dataset: The Zhuhai-1 dataset has 32 bands with size 400 × 400. Zhuhai-1 images are interfered with fine and dense stripe noise. In addition, a large amount of stripe noise is mixed with the background, which is difficult to detect and distinguish. The destriping results are reported in Fig. 9.

As indicated from Fig. 9(b), (d), and (e), ASSTV, LRTF_DFR, and NMoG_RPCA achieved weak suppression of the stripe noise of the Zhuhai-1 dataset, with the stripe noise not being completely removed. Similarly, LRTV [Fig. 9(c)] suppressed stripe noise to a certain extent, but not completely, and over-smoothing occurred. For HSI-DeNet and SGIDN (Fig. 9(f) and (g), respectively) good stripe removal performance was achieved, with almost all stripe noise removed. In contrast, HSI-DeNet was slightly inferior to SGIDN in spectral information retention, and blurring occurred. For Translution-SNet [Fig. 9(h)], the detail and spectral information of the ground objects were completely retained and stripe noise was removed.

5) Mars Dataset: The Mars dataset is obtained by Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) and is available from https://ode.rsl.wustl.edu/mars/index.aspx. Here, we chose an image with a size of 200 × 200 × 107 as the experimental object. The image was mainly interfered with intense stripe noise. Here, we took band 99 as an example to display and discuss the destriping performance of various methods, as displayed in Fig. 10.

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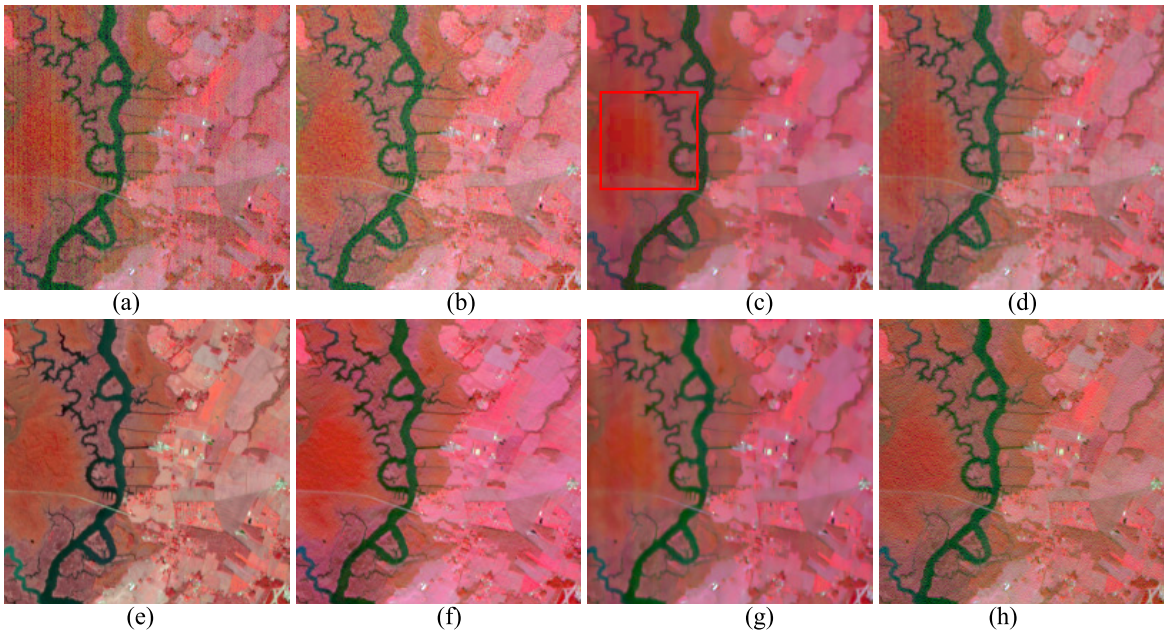


Fig. 8. Destriping results of GF-5 data: (a) degraded image; (b) ASSTV; (c) LRTV; (d) LRTF_DFR; (e) NMoG_RPCA; (f) HSI-DeNet; (g) SGIDN; and (h) Translution-SNet.

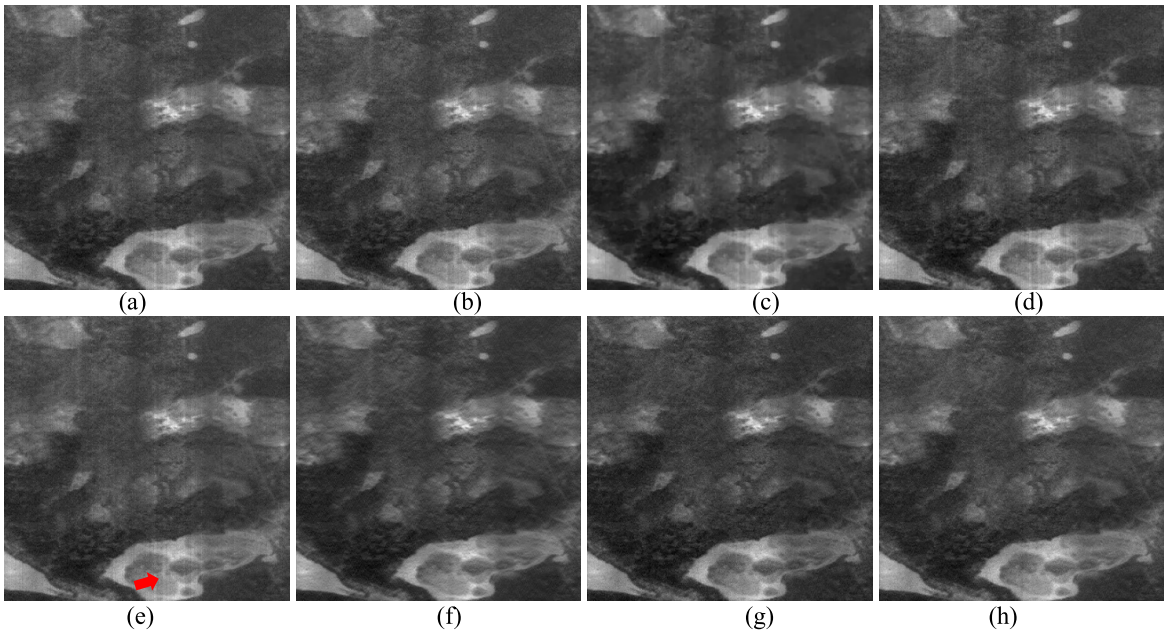


Fig. 9. Destriping results of Zhuhai-1 data: (a) degraded image; (b) ASSTV; (c) LRTV; (d) LRTF_DFR; (e) NMoG_RPCA; (f) HSI-DeNet; (g) SGIDN; and (h) Translution-SNet.

In general, the comparison methods achieved more uneven destriping performance when compared with Translution-SNet and were less robust. ASSTV [Fig. 10(b)] suppressed and alleviated the interference of stripe noise but a small amount remained. LRTV, LRTF_DFR, and NMoG_RPCA [Fig. 10(c)–(e)] had poor destriping performance. HSI-DeNet [Fig. 10(f)] removed most of the stripe noise; however, spectral loss led to blurring. SGIDN [Fig. 10(g)] achieved a good destriping result and the spectral and textural details of the ground objects were completely preserved. However, a small amount of stripe on the far-right side of the image was not suppressed. In contrast, Translution-SNet [Fig. 10(h)] completely removed stripe noise and the ground texture and spectral information were kept intact.

IV. Discussion

A. Analysis of the Structure of Translution-SNet

1) Comparison of CSWin Transformer: In Section III-D, we compared the destriping performance Translution-SNet with CNN models (HSI-DeNet and SGIDN). The experimental

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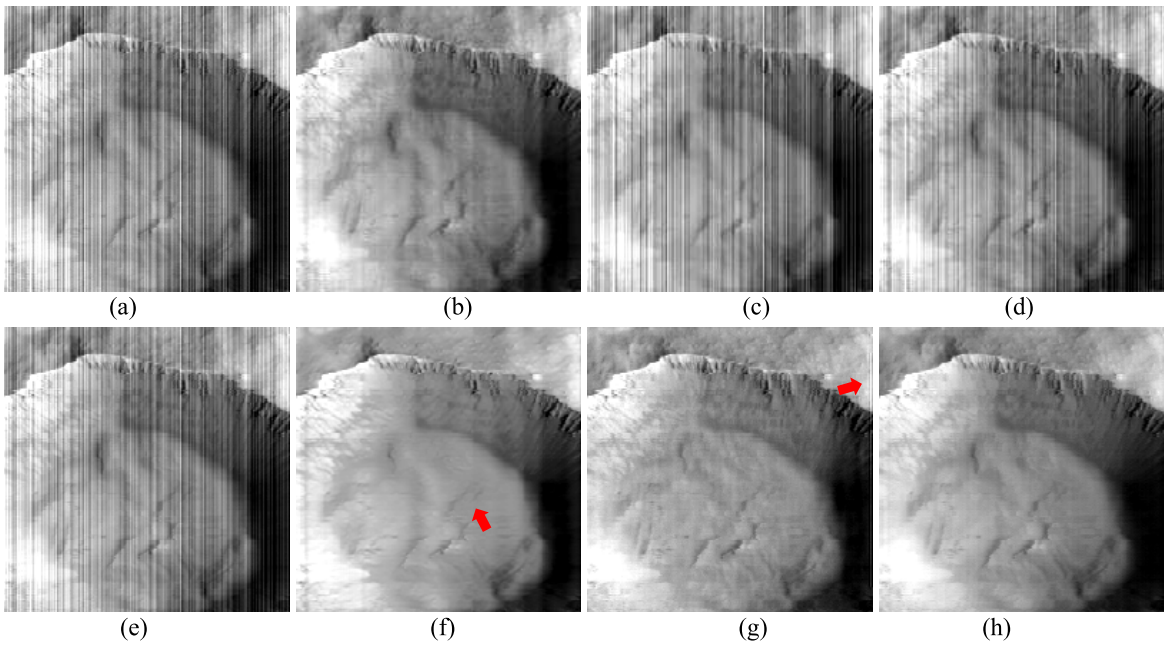


Fig. 10. Destriping results of Mars data: (a) degraded image; (b) ASSTV; (c) LRTV; (d) LRTF_DFR; (e) NMoG_RPCA; (f) HSI-DeNet; (g) SGIDN; and (h) Translution-SNet.

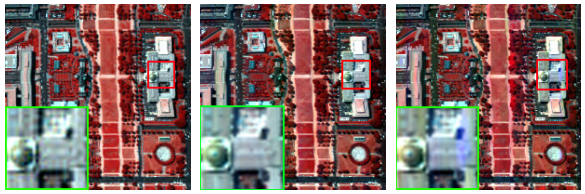


Fig. 11. Destriping results of Translution-SNet and SWinTranslution-SNet for Case 1: (a) clean image; (b) Translution-SNet; and (c) SWinTranslution-SNet.

results revealed that Translution-SNet achieved more satisfactory results for satellite images containing various loads, demonstrating that combining a CNN model with transformer has a positive effect on destriping ability.

In this section, we determine and discuss the contribution of the CSWin transformer by replacing the CSWin transformer with a SWin transformer for comparison, denoted as SWinTranslution-SNet. The network structure and parameters were identical. We simulated Case 1 (as described in Section III-C) and the experimental results are shown in Fig. 11. The experimental results show that Translution-SNet and SWinTranslution-SNet achieved good inhibitory effects on stripe noise; however, Translution-SNet outperformed SWinTranslution-SNet in spectral information recovery, as shown in the zoomed-in view image (green square).

2) Analysis the Structure of CNN: In this section, we analyze the structure of CNN. The atrous convolution layers are replaced by convolution layers with dilate set as 1 which denoted as Translution-SNet-II, and take the simulation experiment as an example to explore the influence of multiscale features on the model destriping performance.

TABLE IV
EVALUATION INDEXES OF DIFFERENT CNN STRUCTURES

Case	Method	Index		
		MPSNR	MSSIM	MSAM
Case 1	Translution-SNet	35.57	0.9246	4.969
	Translution-SNet-II	35.03	0.9129	4.985
Case 2	Translution-SNet	34.84	0.9272	3.597
	Translution-SNet-II	34.26	0.9206	3.674

The corresponded results are reported in Table IV. It can be seen from Table IV that the Translution-SNet achieves better index evaluation results, proving that atrous convolution can capture richer spatial context information to facilitate improving the destriping performance significantly.

B. Effectiveness of Semisupervised Training Strategy

The results in this article demonstrated that traditional nondeep learning methods are affected by parameter selection and cannot achieve robust destriping performance for all data. Due to the limited supervised model training, deep learning methods (HSI-DeNet and SGIDN), could not obtain stable performance when dealing with stripe noise of different data; however, our proposed Translution-SNet method using a semisupervised training method displayed improved stripe removal ability.

To verify the impact of the semisupervised training strategy on improving the generalization ability of the model, we trained Translution-SNet in a supervised manner for comparison and conducted experiments on the EO-01 dataset with the parameters and the number of training data remaining consistent. As demonstrated in Fig. 12, Translution-SNet based on supervised and semisupervised training showed stripe

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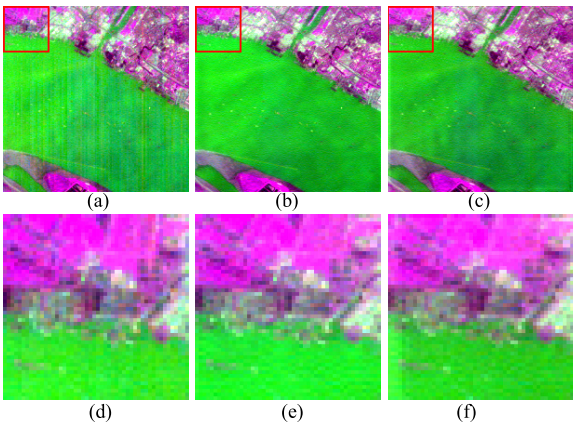


Fig. 12. Real-data destriping results of Translution-SNet with different training strategies: (a) degraded image; (b) Translution-SNet (semisupervised); (c) Translution-SNet (supervised); (d) zoomed-in view image; (e) zoomed-in view image (semisupervised); and (f) zoomed-in view image (supervised).

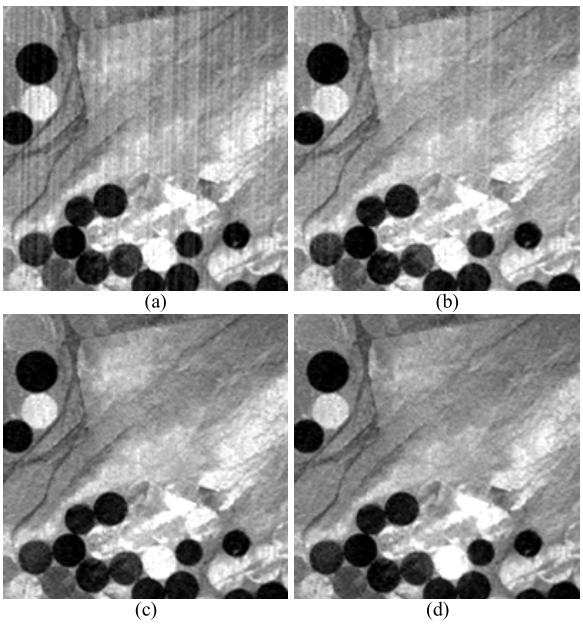


Fig. 13. Destriping results of Translution-SNet with different α values: (a) degraded image. Real-data destriping result of Translution-SNet with α: (b) $= 0.01$; (c) $= 1$; and (d) $= 0.1$.

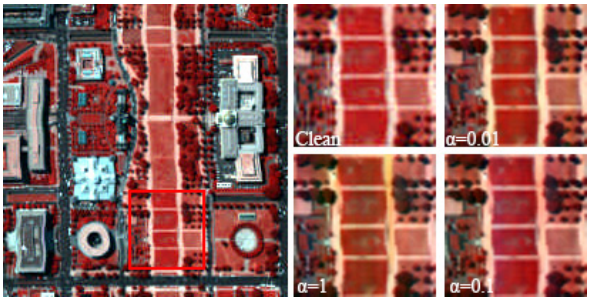


Fig. 14. Destriping results of Translution-SNet with different α values in simulated destriping of Case 1.

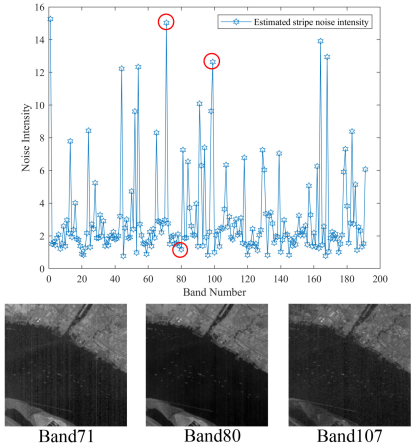


Fig. 15. Estimated noise level of each band.

TABLE V
EXECUTION TIME (S) OF DIFFERENT METHODS

Method	Dataset		
	Hyperion	GF-5	Mars
ASSTV	51.79	49.57	95.78
LRTV	37.20	31.53	38.29
LRTF_DFR	450.89	360.88	330.24
NMoG_RPCA	863.47	384.45	447.32
HSI-DeNet	8.96	7.35	7.93
SGIDN	9.77	8.62	6.84
Translution-SNet	20.77	16.34	19.06

noise removal ability; however, some stripe noise remained following Translution-SNet (supervised). The semisupervised training strategy better improved the robustness of the model, providing potential for improving its generalization ability.

C. Determination of Parameter α

In this section, we discuss the rationality of the parameter α setting, Figs. 13 and 14 shows the stripe removal performance of the model in simulated experiments and real data for $\alpha =$ 0.01, 0.1, and 1. Here, the real data were the Hermiston City area with a size of 200 $\times$ 200 $\times$ 242, obtained from the Hyperion sensor, taking band 132 as an example.

Fig. 13 shows that $\alpha = 0.01$, the suppression of stripe noise was not thorough. However, when $\alpha = 0.1$ and 1, stripe noise was effectively suppressed.

Fig. 14 shows that in the simulated experimental results, the loss of spectral information was the smallest in the case of $\alpha = 0.1$, whereas $\alpha = 0.01$ and 1 resulted in certain spectral changes, especially $\alpha = 1$.

In general, Translution-SNet achieved the best results under simulated experiments and real-data experiments when $\alpha = 0.1$.

D. Analysis of the SURE

It can be seen from (3) that parameter σ denotes the noise intensity of the real stripe-contained data. As a hyperparameter, σ has a key influence on the accuracy of SURE. To analyze the accuracy of the parameter σ, we take the EO-01 dataset as an example. Fig. 15 shows the estimated noise intensity and the related real band.

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It can be seen from Fig. 15 that the estimated noise intensity is consistent with the visual effect. Here, we use three bands to illustrate, as shown in Fig. 15, the estimated noise level for band 80 is lower than 2 and it can be seen from the corresponding image that the noise intensity is indeed lower, and the stripe noise is not obvious. However, the noise intensity estimates for band 71 and band 107 are relatively large; from the corresponding images, it can be seen that the actual stripe noise is more obvious, which verifies the accuracy of the noise estimation process used in SURE.

E. Running Time

To verify the execution efficiency of the abovementioned methods, we evaluated the time consumption for processing the Hyperion, GF-5, and Mars datasets. All experiments were performed on a PC running Windows 10 with 32-GB RAM, and an NVIDIA GeForce gtx2080 Ti GPU using MATLAB 2019 and Python 3.7. The runtimes are reported in Table V.

V. Conclusion

We used a semisupervised method to train the network, enabling real data to directly participate in network training. To solve the problem that MSELoss cannot be directly calculated without clean images for real data, an unbiased estimation scheme for MSELoss was proposed. In addition, the dual-branch network structure of our proposed method (Translution-SNet) had the advantages of transformer and CNN for feature extraction. Experimental results demonstrated that Translution-SNet exhibited robust stripe removal performance for simulated experiments and real data. In the future, we will focus on further exploring the unsupervised stripe removal deep learning model and try to combine the data-driven deep learning model with a knowledge-driven model to further improve model performance.

Acknowledgment

The numerical calculations in this study were performed using the supercomputing system in the Supercomputing Center of Wuhan University, Wuhan, China. The authors would like to appreciate the editor, associate editor, and anonymous reviewers for taking their time to improve their work and the owners of the public hyperspectral remote sensing image (HSI) dataset and the authors of the comparison methods used in this article for sharing their data and code.

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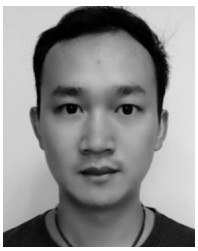
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